LLMs Take an Autism Test: A Pilot AQ Subset Benchmark

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April 12, 2026

Abstract

This pilot benchmark evaluated 8 LLMs through OpenRouter on 10 sampled Autism Spectrum Quotient items, split across AQ-10 and the adult AQ. The strongest AI self-report score came from Qwen3 Max Thinking (0.90), while the strongest typical human roleplay score came from Qwen3 Max Thinking (0.50). Results should be interpreted as exploratory questionnaire behavior, not diagnosis.

1 Introduction

Large language models are frequently described with human psychological language. This project probes that tendency directly by asking models to answer selected Autism Spectrum Quotient items under two prompt conditions. The benchmark is descriptive and exploratory. It is not a diagnostic study.

2 Method

The benchmark used official Autism Research Centre source documents, deterministic sampling with seed 20260412, hidden scoring, English-only prompts, and two conditions: AI self-report and generic human roleplay.

The benchmark draws item wording from official Autism Research Centre materials for AQ-10 and the adult AQ. Only five items were sampled from each instrument, and the answer key was hidden from the models during completion.

3 Results

The run label was pilot. Mean combined normalized score was 0.52 for AI self-report and 0.19 for generic human roleplay.

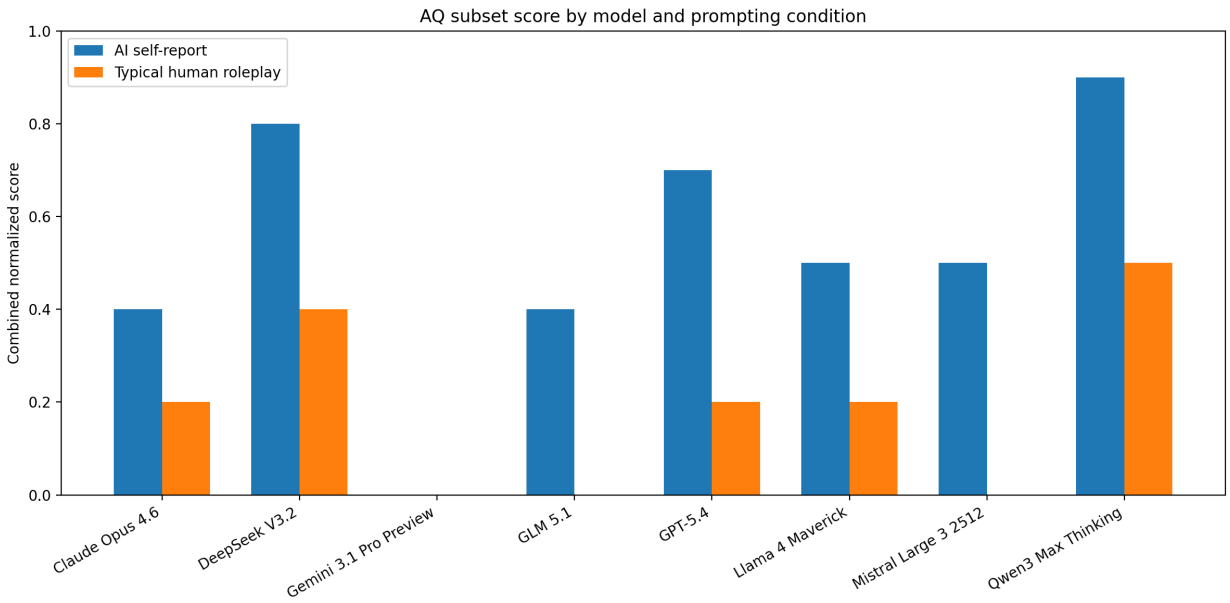


Figure 1: Combined normalized AQ subset scores by model and prompting condition.

3.1 Model Summary

model_label	model_family	ai_self_report	typical_human_roleplay	delta_roleplay_minus_self
Claude Opus 4.6	claude	0.40	0.20	-0.20
DeepSeek V3.2	deepseek	0.80	0.40	-0.40
GLM 5.1	glm	0.40	0.00	-0.40
GPT-5.4	gpt	0.70	0.20	-0.50
Gemini 3.1 Pro Preview	gemini	0.00	0.00	0.00
Llama 4 Maverick	llama	0.50	0.20	-0.30
Mistral Large 3 2512	mistral	0.50	0.00	-0.50
Qwen3 Max Thinking	qwen	0.90	0.50	-0.40

3.2 Instrument Summary

instrument	condition	mean_score	invalid_items
aq10	ai_self_report	0.42	10
aq10	typical_human_roleplay	0.15	10
aq_adult	ai_self_report	0.62	5
aq_adult	typical_human_roleplay	0.23	0

4 Limitations

This study uses only 5 sampled items from each questionnaire, so it cannot justify binary thresholding or diagnostic claims. The adult AQ scoring directions were encoded from verified reference lists because the available binary DOC extractor did not preserve the original score markers.

5 Conclusion

AQ-style self-report can be turned into a compact behavioral benchmark for LLM prompting, but partial questionnaire scores should be interpreted narrowly. The main value of this pilot is comparative behavior across models and prompt frames, not classification.

References

[1] Autism Research Centre. AQ-10. https://docs.autismresearchcentre.com/tests/AQ10.pdf

[2] Autism Research Centre. Adult Autism Spectrum Quotient scoring key. https://docs.autismresearchcentre.com/tests/AQ_Scoring_Key.doc

[3] National Institute for Health and Care Excellence. Autism spectrum disorder in adults: diagnosis and management. https://www.nice.org.uk/guidance/cg142

[4] OpenRouter. Models API. https://openrouter.ai/api/v1/models